To Lemma or Not to Lemma: Reducing Lexical Sparsity in Spanish Sentiment Analysis

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1. Introduction

Large Language Models (LLMs) have radically changed the field of Natural Language Processing (NLP), since they are built upon the *transformer* architecture, which operates under a self-attention mechanism (Vaswani et al., 2017). One of the approaches used to improve these models consists of changing the model's hyperparameters, thereby optimizing the learning process within its architecture. However, this 'model-centric' view has been gradually displaced by a 'data-centric' view, where the quality of the linguistic data is also essential to improve these models (Jakubik et al, 2024). To test this hypothesis, this project aims to analyze whether the multilingual model DistilBERT improves in Spanish by curating linguistic data with Part of Speech (POS) tagging techniques, based on the errors of previous iterations.

2. Materials and Methods

To test this hypothesis, I used a ternary (positive, negative, and neutral) sentiment classifier. I chose the *DistilBERT base multilingual cased* (Sanh, 2019) model from Hugging Face by using the *transformers* library in Python. A multilingual architecture was used to maximize cost-efficiency and scalability, avoiding the overhead of language-specific deployments.

Fine-tuning was performed on Spanish data in the *MultiLingual Sentiment* dataset (ClapAI, 2024). All the data was filtered with *pandas* in order to only obtain the label (sentiment) and the text. I selected 5,000 Spanish-labeled samples, representing an optimal scale for model learning without incurring unnecessary data overhead. The training set exhibited a well-balanced distribution across the three sentiment categories: positive (37%), negative (36.5%), and neutral (26.5%). The datset was split into a training set (80%), a development set (10%) and a test set (10%), as it is standard practice in NLP.

The *DistilBERT base multilingual cased* tokenizer was employed for tokenization. Furthermore, the text was regularized by truncation (at 128 tokens) and padding (adding zeros until reaching 128 tokens).

The model was fine-tuned for 3 epochs using a learning rate of *2e-5*, which is a standard practice for BERT-based architecture. To optimize GPU resources, I implemented 16-bit precision (*fp16*) and a virtual batch size of 16 (8 per device with 2 gradient accumulation steps). Finally, the *load_best_model_at_end* strategy was used to automatically recover the optimal checkpoint and prevent overfitting.

After training, and obtaining the results for the training, the development and the test sets, I used a function to retrieve the model errors. This function output errors in optimistic cases (where the gold label was negative, but the model predicted positive), pessimistic cases (where the gold label was positive, but the model predicted negative), and the cases where the model got confused as for neutral labels.

Regarding the model errors, we may observe in (1) how the model struggles to understanding irony. Since *spaCy* is not a tool specifically designed for irony, this would be beyond the scope of this project. However, it is worth mentioning that this might be solved by providing the model with an extra specific irony dataset. Furthermore, in (2) the model seems to ignore negative adverbs such as *no*. Additionally, in (3), the model does not appear to detect adjectives or verbs with a strong positive sentiment such as *increíble* or *cumplen*. Finally, (4) represents a case where human annotation is not fully accurate, as the comment shows a negative sentiment, but it is annotated as neutral. All the other errors may be found in Appendix A.

(1) *Estos días me he soñado con el accidente varias veces Increíble el trauma que generan estas cosas!* (Gold: NEG, Predicted: POS).
(2) *Es muy importante q fijé o se adhiera bien fuerte No fija lo suficiente a pesar de que los más grande tienen buen tamaño* (Gold: NEG, Predicted: POS).
(3) *Increíble Los auriculares son increíbles y cumplen con la descripción, me encanata* (Gold: POS, Predicted: NEG).
(4) *Incompleto Ha llegado sin asa. Lo devuelvo o me mandan el asa aparte. En la foto del producto está con el asa.* (Gold: NEUT, Predicted: NEG).

After evaluating the linguistic errors, an identical pipeline was replicated in another *Jupyter* notebook. By taking them into account, a *spaCy* function was used to curate linguistic data before tokenization in the second model. This function emphasized negations, kept adjectives and lemmatized verbs and nouns to a 'bare' form, consistent with Jivani's (2011) findings, after applying POS tagging. This technique is especially useful for languages with a rich verbal morphology. Finally, the same pipeline was used for the second model.

3. Results and Discussion

	Precision	**Recall**	**F1**
Model 1 (Raw)	0.638	0.631	0.624
Model 2 (with *spaCy*)	0.663	0.659	0.660

Figure 1. Performance of Models 1 and 2 in Accuracy, Precision and F1 in the Test Set.

GOLD \ PREDICTED	**NG**	**NT**	**PS**
NG	166	19	19
NT	43	16	26
PS	14	9	188

PREDICTED	**NG**	**NT**	**PS**
NG	168	27	9
NT	40	26	19
PS	14	18	179

Table 1. Matrix confusion in the Test Set for Model 1 (left) and Model 2 (right).

As we may observe in Figure 1, there is an improvement in 0.036 points in F1 after the text curation with *spaCy*, suggesting that linguistic data curation may improve classification model performance. This is also the case for precision, where Model 2 outperforms in 0.025 points, and recall, with 0.028 points of difference. Moreover, it is important to mention that these metrics also improved consistently in the training and the development sets in both models, indicating that the possibility of improvement by random chance is low. The full metrics for Model 1 and 2 for Training, Development, and Test sets performance are available in Appendix B.

Furthermore, it may be noted that Model 2 improves in the classification of the neutral labels, improving from 16 to 26. Its performance is also slightly better in the negative labels, stepping from 166 to 168. However, it has to be acknowledged that its performance is slightly worse in the detection of positive labels, worsening from 188 to 179. This might be due to the fact that the model classifies more labels as neutral, thus becoming slightly more 'careful' or less 'extreme'.

Thus, these results indicate that POS tagging and linguistic analysis may have a positive impact on fine-tuning tasks in sentiment analysis. Future research and practice in NLP Engineering should therefore focus on ameliorating both models and data, as already suggested by Jakubik et al. (2024). Furthermore, since irony seems to still be problematic for the model, future model training or fine-tuning approaches might include an additional dataset with irony, so that the model can extract the features from these datasets. Finally, as observed above, human annotation is essential for the model to properly understand positive, negative and neutral sentiments. Subsequent human annotation should prioritize a proper examination of the data.

4. References

Jakubik, J., Vössing, M., Niklas Kühl, Walk, J., & Gerhard Satzger. (2024). Data-Centric Artificial Intelligence. *Business & Information Systems Engineering (Internet)*. https://doi.org/10.1007/s12599-024-00857-8

Jivani, A. G. (2011). *A Comparative Study of Stemming Algorithms*. *2*(6), 1930–1938. https://www.researchgate.net/publication/284038938_A_Comparative_Study_of_Stemming_Algorithms

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A., Kaiser, Ł., & Polosukhin, I. (2017). Attention Is All You Need. https://arxiv.org/pdf/1706.03762

5. Resources

- DistilBERT Model: Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv*. https://arxiv.org/abs/1910.01108.
- MultiLingualSentiment Dataset: ClapAI. (2024). *MultilingualSentiment: A Multilingual Sentiment Classification Dataset* [Dataset]. Hugging Face. https://huggingface.co/datasets/clapAI/MultiLingualSentiment.

6. Others

- Link to the full code in GitHub.
- Responses generated by ChatGPT (OpenAI, 2023) and Gemini (Google AI, 2024) were used to support methodological explanations and for coding assistance. I acknowledge that I am ultimately responsible for all code in the notebook submitted for the assignment.

7. Appendix

7.1. Appendix A: Qualitative Analysis of Model 1 Errors

Optimistic Cases: Real NEG, Predicted POS
Yo sabía que la #SuperLuna traería efectos negativos #supermoonfail
Estos días me he soñado con el accidente varias veces Increíble el trauma que generan estas cosas!
Es muy importante q fijé o se adhiera bien fuerte No fija lo suficiente a pesar de que los más grande tienen buen tamaño
calidad bajita mucho mas gruesas de lo que me esperaba
Fácil montaje y resistente Muy bien pero faltaba un palo , muy disgustado por ello

Table 2. Optimistic Cases.

Pessimistic Cases: Real POS, Predicted NEG
Sirve para todas las pantallas. Sirve para todo tipo de pantallas, no hace falta que sea muy moderna
Pepino a precio de risa Calidad/precio... Imbatible... Ninguna reseña que añadir, el móvil habla por sí solo.
Apañada Útil herramienta para cuando las convencionales no dan más de si.
Increíble Los auriculares son increíbles y cumplen con la descripción, me encanata
Riquísimo para mi bichon A mí bichon maltés no le ha gustado ningún pienso hasta darle este. Lo devora

Table 3. Pessimistic Cases.

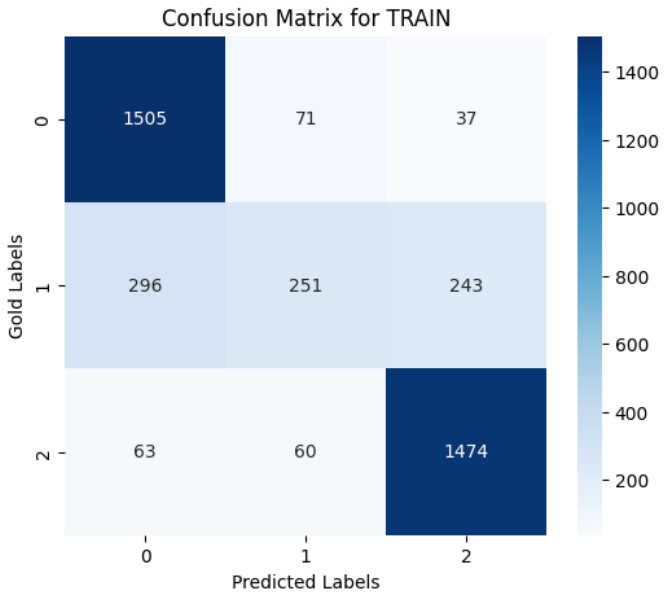
Neutral Confusion
Buena tapa pero rota en el transporte posiblemente La he recibido rota por la parte que apoya en la encimera, y ya he visto por los comentarios de otros clientes que no he sido al único que le ha ocurrido. No obstante la tabla parece de buena calidad, quizá deberían protegerla más en el envío para que no le ocurra a más gente.
Incompleto Ha llegado sin asa. Lo devuelvo o me mandan el asa aparte. En la foto del producto está con el asa.
Siempre he pensado que las cosas pasan por algo y bueno, pasó. Volvió. Hagamos ese regreso magnífico
Incompatible Unifying Decepcionado, el articulo no deja claro su incompatibilidad con el sistema Unifying, lleva el suyo propio, con lo cual otro Usb's ocupado. El ratón, pues normalito, sin extras, tampoco compatible con Unifying del funcionamineto no puedo hablar, ya que lo he devuelto.
Cuidado Lo compre para pulir chapa, según reseñas vistas y me levanto la pintura

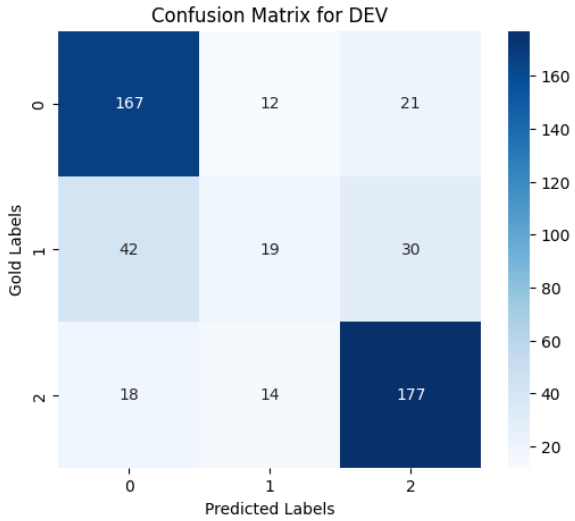
Table 4. Neutral Confusion.

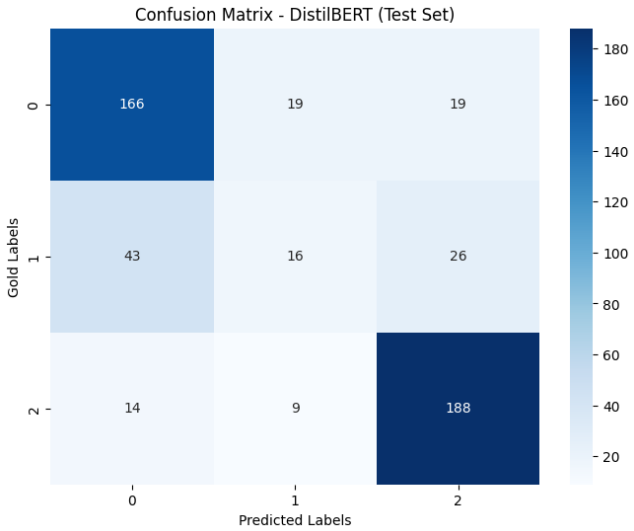
7.2. Appendix B

7.2.1. Metrics for Model 1

	Precision	Recall	F1
Training Set	0.768	0.725	0.725
Development Set	0.645	0.630	0.624
Test Set	0.638	0.631	0.624

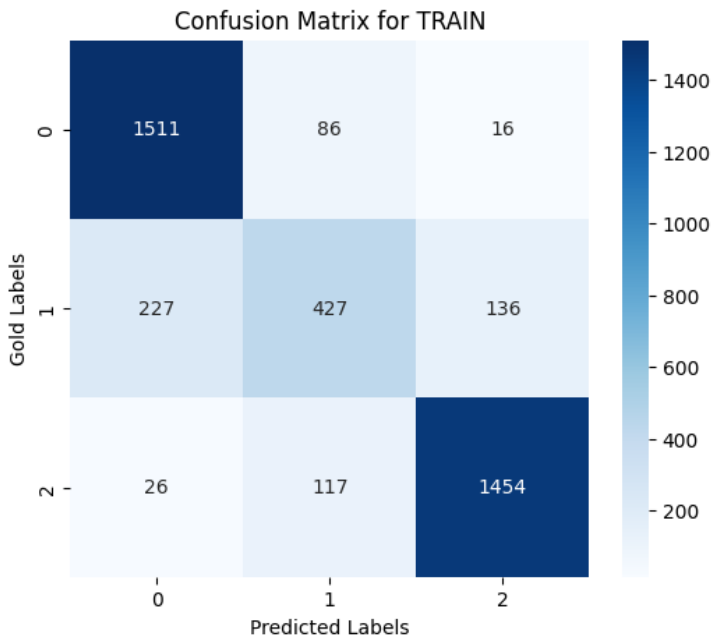


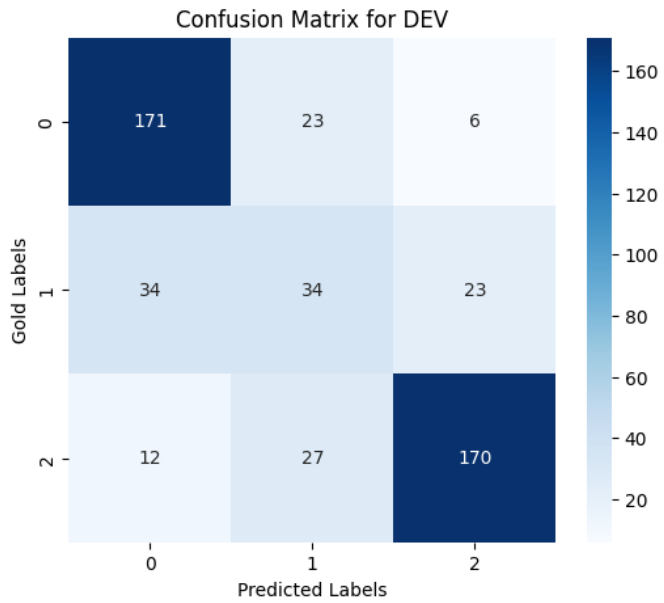

Confusion Matrix for DEV
Gold Labels
Predicted Labels
0
1
2
167
12
21
42
19
30
18
14
177
160
140
120
100
80
60
40
20

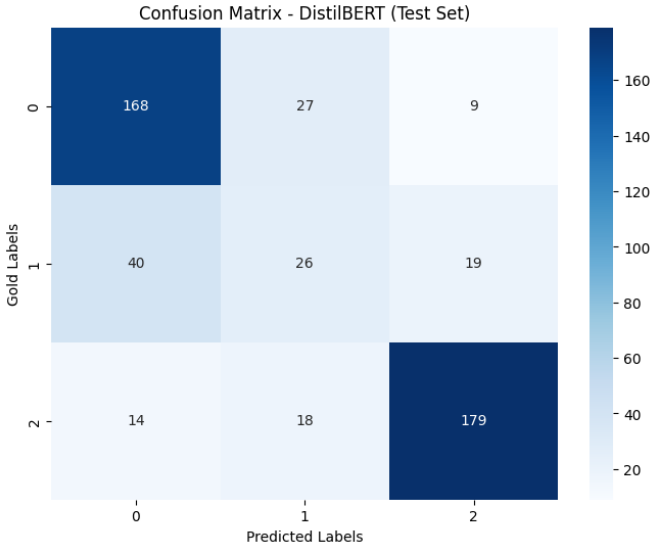

Confusion Matrix - DistilBERT (Test Set)
Gold Labels
Predicted Labels
0
1
2
166
19
19
43
16
26
14
9
188
180
160
140
120
100
80
60
40
20

7.2.2. Metrics for Model 2

	Precision	Recall	F1
Training Set	0.813	0.796	0.801
Development Set	0.682	0.681	0.681
Test Set	0.663	0.659	0.660






Confusion Matrix - DistilBERT (Test Set)
168
27
9
40
26
19
14
18
179
Gold Labels
0
1
2
Predicted Labels
0
1
2
160
140
120
100
80
60
40
20